

MWD Survey Simulation

Taking a downhole survey, step by step, on a simulated rig floor

⚠ FOR UNDERSTANDING PURPOSES ONLY

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Flutter / Dart

Hoist & slack

Pump control

Survey cycle

Strip chart

1. What It Is

A training simulation of the rig-floor routine for taking an **MWD (Measurement While Drilling) survey** — the reading of the well's direction while drilling. The driller moves the string, controls the pumps and follows on-screen prompts, and the downhole tool “reports” back the azimuth, inclination and magnetic dip. It is reached from Operations › LWD / MWD › MWD & DD › Survey.

It began as a desktop training simulator and was re-created in Flutter, keeping the same physical behaviour.

2. The Two Pages

The screen has two pages that you swipe between.

Page	What it shows
Main page	Controls, pump and pressure, the survey downlink card and the live pulse display, with a banner at the top telling you what to do next
Strip chart	Three traces plotted against time: block position, bit depth and pump pressure

3. The Controls

- **Hoist and Slack** — raise or lower the string; the block moves, the depth changes and the weight on hook and weight on slips update.
- **Pump toggle and MP 1** — turn the mud pump on and off. Standpipe pressure, annular pressure losses and total strokes follow.
- **Draw Works and Hook Load** gauges, and a depth read-out.
- **Reset** returns everything to the start.

4. The Survey Cycle

A survey is a fixed sequence, guided by a coloured banner:

1. **“Time for survey!!”** — the banner flashes red to start the cycle.
2. **“Release the torque”** — the driller hoists and slacks the string to the right position.
3. **“Turn off the pump”** — the pump is stopped.
4. **One-minute pump-off wait** — a countdown runs while the tool takes its reading. A fast-forward button shortens it, and the pump cannot be restarted until it finishes.
5. **“Turn on the pump”** — the pump is restarted, which sends the survey signal up as a mud pulse.
6. **“Survey received”** — the azimuth, inclination and magnetic dip appear above the pulse display for a couple of seconds and then fade.

5. The Survey Downlink Card

A card pairs the tool’s “XYXX” downlink with a list of steps and parameters. As each step of the procedure is completed it turns green, so the learner can see how far through the downlink sequence they are, and the tool receiver value switches to “XYXX” when the survey is acknowledged.

6. Why It Matters

Learning the rhythm

Surveys are routine on a drilling rig, but the order matters: release torque, stop the pump, wait, restart. The simulator lets someone rehearse the timing without any real equipment, with the banner reminding them of the next action.